Global Token Dictionary Augmented Patch Transformer for Hyperspectral Image Reconstruction

Yubao Sun, Hongkang Luan, Zhenyu Zhang, Le Sun, and Qingshan Liu, *Senior Member, IEEE*

***Abstract*—Reconstructing hyperspectral images (HSIs) from red, green, and blue (RGB) images is a cost-effective and convenient method to obtain fine spatial–spectral information. The reconstruction methods based on Transformers are currently very competitive. They mainly take each pixel as the token to calculate the correlation within the spatial window, or take each spectral band as the token to calculate the correlation along the entire spectral dimension. Although they have improved the reconstruction quality, they still have limitations in learning global contextual information while preserving local spatial–spectral features. To address these issues, we propose a novel global token dictionary augmented patch Transformer (GAP-Trans) network to learn the inverse reconstruction mapping. The core innovation of GAP-Trans is to use multiscale local patches as the basic unit to calculate spectral-wise mixed attention (SMA) within local patches and between local patches and global contextual information generated by the global token dictionary generation module. Specifically, GAP-Trans is configured with dual branch to calculate the cross-attention between the global token dictionary and local patches, as well as the self-attention within local patches. The fast Fourier fusion operator is then adopted to fuse the outputs of the dual branch in the frequency domain. This mechanism enables interaction between global and local information in an effective way, promoting cross-scale spatial and spectral feature fusion. Furthermore, we construct the encoder–decoder network based on GAP-Trans to obtain the final reconstruction. Extensive experiments demonstrate that our proposed network significantly outperforms baseline methods in terms of reconstruction accuracy.**

***Index Terms*—Global token dictionary, hyperspectral image (HSI) reconstruction, patch Transformer, spectral-wise mixed attention (SMA).**

I. Introduction

HYPERSPECTRAL imaging is a revolutionary imaging technique that has finer spectral resolution than conventional imaging systems. Unlike traditional cameras that only record red, green, and blue (RGB) information, hyperspectral imaging captures a large number of spectral bands spanning a wide range of wavelengths. The vectorized spectral response values of each pixel in hyperspectral images (HSIs) form a spectral signature [1]. This rich spectral information can unlock a hidden world, revealing specific attributes of targets in the scene. HSIs have many applications in diverse fields, such as remote sensing for monitoring environmental changes, precision agriculture for optimizing crop health, and medical imaging for disease diagnosis [2], [3], [4]. However, in order to capture spatial–spectral data cubes, current HSIs typically require time-consuming scanning operations in either spatial or spectral dimensions. Meanwhile, obtaining high spectral resolution often comes at the cost of reducing spatial resolution.

Directly reconstructing HSI from high-resolution RGB images is an appealing alternative that enables faster and more cost-effective acquisition of spectral information. The three channels of an RGB image sample spectral information at wide intervals, which can be seen as a mixture of multiple narrow spectral bands in an HSI. This inherent relationship allows the extraction of fine-grained spectral data from RGB images through spectral reconstruction [5].

In recent years, deep learning-based methods have been widely applied to HSI reconstruction research. As a representative deep learning model, convolutional neural networks (CNNs) have achieved superior performance compared with traditional methods based on prior regularization. However, due to the size limitation of convolutional kernels, they typically have relatively small receptive fields, which limits their ability to capture long-range dependencies and self-similarity of spatial–spectral features [6], [7]. With the remarkable progress of Transformers in computer vision, researchers have begun to apply a visual Transformers to image reconstruction tasks. Thanks to the advantages of self-attention mechanism, Transformer-based methods can handle long-distance dependencies and nonlocal self-similarity and have better reconstruction quality compared to CNN-based methods. There are currently two main design approaches for Transformer self-attention computation used for HSI reconstruction. The first approach takes the vectorized spectral

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response values of each spatial pixel as a token and computes self-correlation within a local spatial window. In order to capture self-correlation over a large scale, this approach usually enlarges the local spatial window's size. However, due to the quadratic computational complexity, it incurs high time costs. The second approach takes each spectral band, with its full spatial extent, as a token and calculates the correlation across different spectral bands. While this reduces computational complexity, it overlooks the contribution of local features within the spatial dimension to reconstruction quality. How to efficiently capture global contextual information while preserving local details remains a key challenge for improving reconstruction performance.

In order to cope with this challenge, we design a novel global token dictionary augmented patch Transformer (GAP-Trans) and use it to construct an encoder and decoder network GAP-Trans network (GAPT-Net) for HSI reconstruction. GAP-Trans achieves the fusion of global contextual information and local spatial–spectral features by utilizing spectral-wise mixed attention (SMA) within local patches and between local patches and the global token dictionary in a dual-branch framework. Specifically, the global token dictionary generation module is employed to aggregate the global information into the global token dictionary. The dual branch of GAP-Trans treats the global token dictionary as a shared value and query respectively, and calculates self-attention and cross-attention along the spectral dimension with all local patches. In both branches, the coefficient matrices are used as weights to linearly combine the corresponding value matrix terms. Additionally, we design a fast Fourier fusion operator to transform the output of the dual branch into the frequency domain for weighted fusion, achieving a balance between global and local features to enhance reconstruction quality. Extensive experimental validations demonstrate that the proposed reconstruction network significantly improves the quality of reconstructed HSIs with delicate spatial structures and accurate spectral signatures. The ablation studies further verify the effectiveness of the designed network modules. The main contributions of this article are summarized as follows.

1) We propose a novel Transformer, GAP-Trans, designed specifically for multiscale local patches. It treats the spectral bands within these patches as tokens to compute spectral-wise attention, effectively reducing computational complexity while simultaneously extracting coupled spatial and spectral features to enhance reconstruction performance.
2) We design an SMA mechanism that combines cross-attention and self-attention within a dual-branch framework. The cross-attention calculates the correlation between the global token dictionary and all local patches, thereby integrating global contextual information into each local patch and facilitating the fusion of long-range global features with fine local details.
3) We further introduce a fast Fourier fusion operator to perform weighted fusion of the outputs from cross-attention and self-attention in the frequency domain. This enables the adaptive integration of low-frequency global contextual information with high-frequency local features, thereby improving reconstruction quality.

II. Related Work

A. Hyperspectral Imaging Technology

Hyperspectral imaging technology represents a significant advancement in capturing and processing electromagnetic spectrum data [8]. Unlike traditional imaging systems that capture only a limited number of spectral bands, hyperspectral imaging senses the 2-D scene across numerous bands with fine spectral resolution, enabling detailed analysis of the objects in the image. The acquired 3-D data cue serves as the basis for precise material and environmental analysis across various applications, including remote sensing, agriculture, environmental monitoring, and medical imaging. Hyperspectral imaging systems typically use a 2-D sensor, and to capture the full spatial–spectral data cube, they are paired with scanning devices such as push-broom scanners, whisk-broom scanners, or full-aperture scanners [9], [10], [11]. As a result, HSIs are often bulky and costly. Additionally, due to the limited efficiency of light usage during exposure, achieving high spatial and spectral resolution simultaneously is challenging, with improving spectral resolution often coming at the expense of spatial resolution. Consequently, pursuing a way to economically and rapidly acquire HSIs has emerged as a prominent research focus.

B. HSI Reconstruction Methods

Reconstructing HSIs from RGB images offers a cost-effective and efficient method for acquiring hyperspectral data cubes. To overcome the hardware limitations associated with hyperspectral imaging systems and to reduce the need for specialized expertise, the development of algorithms capable of reconstructing HSIs from easily accessible RGB images holds paramount importance for both research and practical applications. Early reconstruction algorithms primarily relied on prior knowledge of HSIs to build reconstruction models, including sparse coding, low-rank representation, and dictionary learning models [12], [13], [14]. However, these methods typically require iterative optimization to estimate sparse representation coefficients, making the reconstruction process computationally expensive. In addition, their representation ability is relatively limited and their generalization performance is often poor, which affects the quality of reconstruction.

In recent years, many deep learning-based methods have emerged in computer vision tasks [15], [16], demonstrating commendable performance. Xiong et al. [17] introduced the deep learning network HSCNN, leveraging CNNs as its foundation to learn the inverse reconstruction mapping. Addressing the challenge of requiring an explicit spectral response function for upsampling within HSCNN, Shi et al. [18] proposed a simple convolutional layer as a substitute for manual upsampling operations. Furthermore, subsequent advancements replaced the conventional convolutional layers in HSCNN with residual and dense modules, leading to the development of the HSCNN-R and HSCNN-D models.

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These enhancements substantially improved the reconstruction quality, albeit at the cost of a proportional increase in computational complexity. Stiebel et al. [19] innovatively combined spectral reconstruction and semantic segmentation tasks by employing the widely recognized U-Net encoder–decoder architecture. This approach encompassed spectral information at the pixel level and incorporated contextual information from neighboring pixels, thereby enhancing overall reconstruction integrity and outperforming previous methods based on single-pixel independent learning. In addition, Li et al. [20] incorporated prior knowledge of camera spectral sensitivity and the interdependencies among intermediate features into the design of a deep reconstruction network, leading to further improvements in reconstruction quality.

While CNN-based deep learning methods have achieved competitive reconstruction results, the limited receptive field of convolutional kernels constrains their ability to capture long-range correlations across both spatial and spectral dimensions. In light of this, the growing success of Transformers in natural language processing and computer vision has motivated their gradual adoption in HSI reconstruction tasks, owing to their strong capability in modeling global dependencies.

C. Vision Transformer

The Transformer architecture [21] was originally proposed for machine translation tasks and has substantially advanced natural language processing by effectively modeling long-range dependencies [22]. A notable milestone in this line of research is the Vision Transformer (ViT) introduced by Han et al. [23], which departs from conventional CNN-based designs and applies a pure Transformer architecture to image classification tasks. It inherits the self-attention design from the original Transformer, effectively capturing long-range correlations of visual features, and also brings key advancements. At present, the Transformer framework has been widely applied in various image-related tasks, such as image classification, object detection, image segmentation, denoising, and image super-resolution [24], [25], [26], [27], [28].

Cai et al. [29] constructed a mask-guided spectral-wise Transformer (MST) network based on the principle of compressive sensing, which is a pioneer in using Transformers for HSI reconstruction. Their approach treated each spectral band as a token and employed self-attention along the spectral dimension, leading to notable performance enhancements over CNN-based models. Building upon the self-attention mechanism along the spectral dimension, Luan et al. [30] introduced MS^{3}T and emphasized the representation of the interrelationship between spatial and spectral dimensions to elevate reconstruction quality significantly. Additionally, Cai et al. [29] extended their MST network and proposed the MST++ network based on U-Net, achieving commendable reconstruction results. Yao et al. [31] proposed a network called SPECAT utilizing a cumulative-attention mechanism within an efficient hierarchical framework to learn nonlocal mutation features. By jointly refining spatial details and spectral fidelity in a coherent manner, SPECAT achieved good performance while maintaining low computational cost in high-resolution HSI reconstruction tasks. Zhang et al. [32] proposed a network, CASSNet, which leverages a cross-attention enhanced spectral–spatial interaction mechanism to achieve a synergistic improvement in both spectral fidelity and spatial resolution for HSI super-resolution. By employing a vision-language model (VLM)-guided prompt learning, Lee et al. [33] proposed a single model PromptHSI to deal with multiple degradations of HSIs. Wu et al. [34] further proposed a novel multiprompt frame network named MP-HSIR, which can effectively integrate spectral, textual and visual prompts to achieve universal HSI restoration across diverse degradation types and intensities. Xu et al. [35] designed a sparse spectral Transformer that enables adaptive masking of nonsignificant details, thereby enhancing reconstruction accuracy. Moreover, Transformer has also been applied in the task of unmixing HSI. For example, Ghosh et al. [36] proposed a Transformer-based network for hyperspectral unmixing, further demonstrating the advantages of Transformer in spectral dimension modeling.

Although Transformer networks has boosted the hyperspectral reconstruction quality, they still suffer from the issue of losing local spatial features during learning global contextual information [37], [38]. In order to further improve the reconstruction quality, it is necessary to design a novel type of Transformer that can learn both global and local features simultaneously.

III. GAP-Trans Network

A. Overview

Fig. 1 shows the overall architecture of the proposed GAPT-Net. The proposed network takes the RGB image $I_{RGB} \in R^{H\times W\times 3}$ as input, where H, W, and 3 are the height, width, and number of channels, respectively. Its main body adopts GAP-Trans as the basis element to construct the encoder–decoder architecture. Through the upsampling and downsampling operations of the encoder–decoder and the multiscale patch division of GAP-Trans, it effectively captures the spatial–spectral features of different scales and promotes the recovery of fine structures.

Specifically, we first utilize a shallow information extraction module (shallow extraction) to extract initial features and adjust the number of feature map channels. Subsequently, in the encoding and decoding stages, we design a multiscale patch Transformer block, which is composed of three GAP-Trans with different local patch sizes and three corresponding global token dictionary generation modules (GTDGs). The receptive fields of the three Transformers in the Transformer block range from 16×16, 8×8, to 4×4. The design of patch sizes from large to small in the Transformer block helps the reconstruction network prioritize long-range contextual information and then extract local feature information. At the same time, the three levels of downsampling operations in the encoder further increase the coverage of the receptive fields, enabling the extraction of spatial–spectral features at large scales. Furthermore, we use residual links to connect the corresponding processing blocks between the encoding and decoding stages so that complementary information is transferred from the encoding stage to the decoding stage, which helps the decoding stage reconstruct the image better

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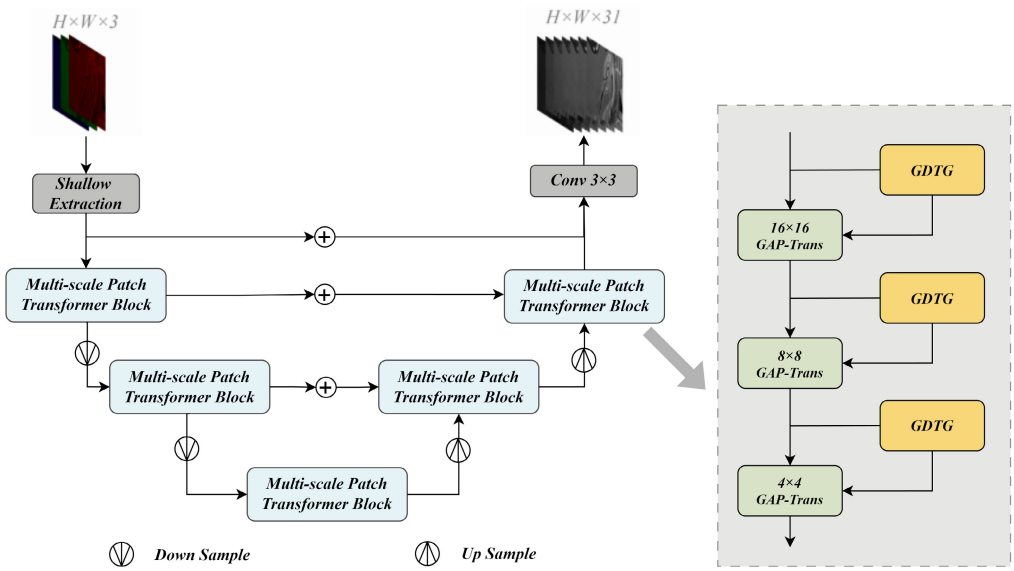


Fig. 1. Architecture of GAP-Trans network for HSI reconstruction.

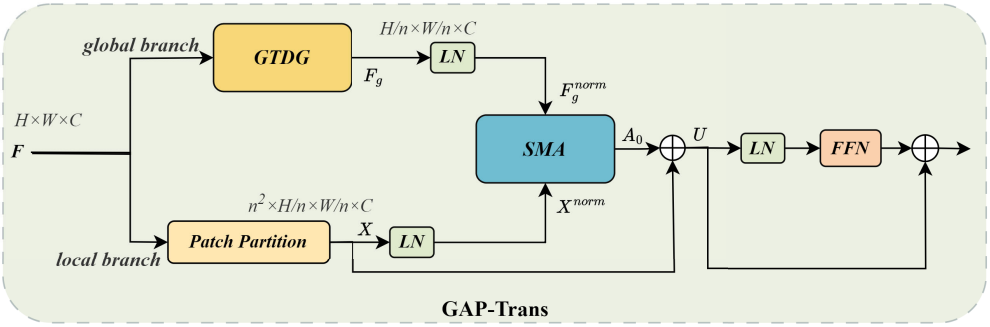


Fig. 2. Diagram of GAP-Trans. SMA corresponds to the calculation of spectral-wise mixed attention.

and reduces information loss. Finally, we use a 3×3 convolutional layer to adjust the channel number of the output of the decoding stage to 31.

B. Global Token Dictionary Augmented Patch Transformer

Fig. 2 illustrates the architecture of the GAP-Trans. To simultaneously extract features along the spectral dimension and the spatial dimension, we divide the input feature maps along the spatial dimension and perform SMA calculation under the guidance of a global token dictionary. By this operation, GAP-Trans is capable of injecting global contextual information into the local patches, thereby facilitating better reconstruction of the features of each patch.

Specifically, the GAP-Trans takes feature map $F \in R^{H \times W \times C}$ as input and processes it with a dual branch. The local branch divides the input feature map F into n^2 nonoverlapping local patch matrices of spatial size $M \times M$ by the patch partition operation. We denote them as $X \in R^{n^2 \times M \times M \times C}$, where $n^2 = HW/M^2$.

Another global branch assembles global contextual information into a global token dictionary $F_g \in R^{M \times M \times C}$ by means of the global token dictionary generation module. Then, we normalize the global token dictionary F_g and all the local window matrices X using LayerNorm [39], which can be formulated as

$$F_g^{norm} = LayerNorm\left(F_g\right) \tag{1}$$

$$X^{norm} = LayerNorm\left(PatchOperation\left(F\right)\right). \tag{2}$$

Next, the normalized results flow into the SMA, and its output can be calculated as

$$U = SMA\left(F_g^{norm}, X^{norm}\right) + X. \tag{3}$$

The detailed computation flow of SMA, will be introduced in Section III-D.

Previous works have demonstrated that the feed-forward layer in the standard Visual Transformer has limited capability in utilizing local contextual information. To address this problem, we add a depth-wise convolutional module to the feed-forward network (FFN) layer. Specifically, we first use a convolutional layer with a kernel size of 1×1 to increase the feature dimensionality. Then, a depth-wise convolution module with a kernel size of 3×3 captures local contextual

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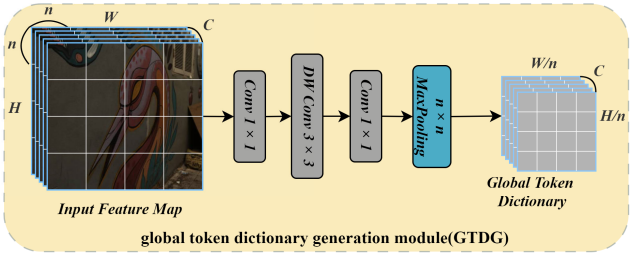


Fig. 3. Architecture diagram of global token dictionary generation module.

information. Finally, another convolutional layer with a kernel size of 1×1 is used to reduce the feature dimensions to be the same as the original input. We use GELU as the activation function after each convolution operation. The calculation process of FFN can be expressed as

$$Z = Conv_{1\times1}\left(GELU\left(DWConv_{3\times3}\left(Conv_{1\times1}(LayerNorm(U))\right)\right)\right). \tag{4}$$

C. Global Token Dictionary Generation Module

Fig. 3 shows the architecture diagram of the global token dictionary generation module. It first extracts spatial–spectral features from the input feature map through multiple convolution operations, and then uses max-pooling operation to collect global contextual information to obtain a global token dictionary. Meanwhile, the max-pooling operation reduces the spatial dimension of the feature map, enabling it to match the size of local patches in GAP-Trans.

Specifically, we initially use a convolutional layer with a kernel size of 1×1 to boost feature dimensionality along the spectral dimension. Then, a 3×3 depth-wise convolution module focuses on the extraction of spatial features. Then, another 1×1 convolutional layer is used to restore the feature dimension to the input state. Finally, an $n \times n$ 2-D max-pooling layer is utilized to compress the spatial size of the global token dictionary to match that of the local window currently being processed. We use GELU as an activation function after each convolution operation.

D. Spectral-Wise Mixed Attention

The calculation of SMA is the core of GAP-Trans. As shown in Fig. 4, SMA takes the spectral band of multiscale local patches as tokens and calculates SMA between local patches and the global token dictionary in a dual branch. The calculation formulation of SMA is defined as

$$SMA\left(F_g^{norm}, X^{norm}\right) = F_u\left(A_s\left(F_g^{norm}, X^{norm}\right), A_c\left(F_g^{norm}, X^{norm}\right)\right) \tag{5}$$

where A_c denotes the cross-attention calculation, A_s denotes the self-attention calculation, and F_u denotes the fusion operation in the Fourier domain.

The global branch takes F_g^{norm} as the value and X^{norm} as query and key matrix and calculates the spectral-wise self-attention within each local patch. Then, the obtained coefficient matrix is used as weights to linearly combine the terms in the value matrix to obtain the updated output. In detail, we first reshape the dimension size of global token dictionary F_g^{norm} into $1 \times C \times M^2$ and name it as $D_{global} \in R^{1\times C\times M^2}$, which has global contextual information. Then, a convolutional feature extraction process is applied to the local patch matrix X^{norm}. This operation integrates the local modeling advantages of convolutional operations into the attention calculation, thereby enhancing the network's ability to capture local details. In detail, X^{norm} first undergoes a 1×1 convolutional layer to map features and double its feature dimension. Then, a 3×3 depth-wise convolutional module is used for feature extraction. The result is subjected to uniform chunking along the spectral dimension and reshaped to the size of $C \times M^2$. Finally, all local window features are stacked to obtain keys $K_s \in R^{n^2\times C\times M^2}$ and querys $Q_s \in R^{n^2\times C\times M^2}$. The above projection operations are defined as

$$[Q_s, K_s] = Stack\left(Chunk_C^2\left(DWConv_{3\times3}\left(Conv_{1\times1}\left(X^{norm}\right)\right)\right)\right) \tag{6}$$

where $Conv_{1\times1}(*)$ and $DWConv_{3\times3}(*)$ denote the convolution operation with a convolution kernel size of 1×1 and the depth-wise convolution operation with a convolution kernel size of 3×3, respectively. $Stack(*)$ symbol represents a stack operation. $Chunk_C^2(*)$ represents splitting a tensor into a specified number of chunks along the given dimension, the superscript represents how many chunks are divided, and the subscript represents which dimension is divided. Then, we calculate the self-attention mechanism as

$$A_s\left(D_{global}, X^{norm}\right) = SoftMax\left(\sigma_1 Q_s K_s^T\right) D_{global} \tag{7}$$

where σ_1 is a learnable parameter. Considering the variation in spectral intensity at different wavelengths, we use σ_1 to adaptively adjust the weights of correlation coefficients. The $Q_s K_s^T$ operation treats all spatial dimension features of a single spectral band as a token and performs attention calculation along the spectral dimension. The resulting attention matrix has the size of $C \times C$. This approach of attention calculation can significantly reduce computational complexity. Additionally, the local patch strategy allows for the simultaneous extraction of spatial features while speeding up computation, thus accelerating model training.

As described earlier, D_{global}, which contains global information, its tensor shape has a single dimension in the first dimension (with a size of 1), while K_s and Q_s, which contain local information, have a size of n^2 in the first dimension (batch dimension). However, due to PyTorch's broadcasting mechanism, the missing dimension in D_{global} during matrix computation can be automatically adjusted to match the dimensions of K_s and V_s. Specifically, $1 \times C \times M^2$ is replicated as $n^2 \times C \times M^2$, enabling matrix computation between them.

Furthermore, to further enhance the fusion of local and global features, the local branch takes a new fusion manner dual to the global branch. The global token dictionary D_{global} is taken as query matrix. Local patches are first projected as key matrix K_c and value matrix V_c. The spectral-wise cross-attention between the global token dictionary and local patches is performed to obtain the coefficient matrix. This allows us to extract local features under the guidance of global contextual

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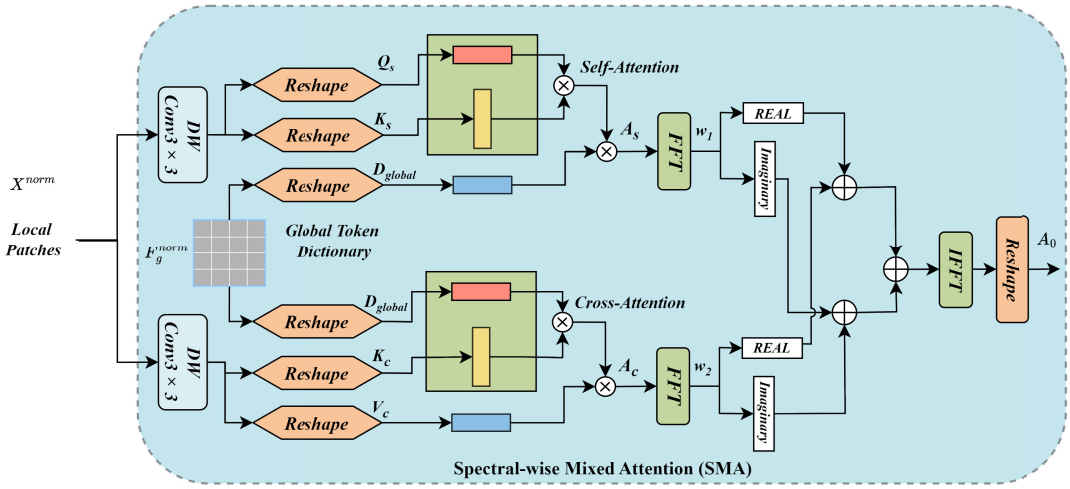


Fig. 4. Architecture diagram of SMA composed of self-attention and cross-attention.

feature. Similarly, the coefficient matrix is used as the linear combination weight of the terms in the value matrix. The calculation of this self-attention can be formulated as

$$A_c\left(D_{global}, X^{norm}\right) = SoftMax\left(\sigma_2 D_{global} K_c^T\right) V_c. \quad (8)$$

Next, we transform A_c and A_s into the frequency domain using fast Fourier transform (FFT) for weighted addition processing. Then, we transform the weighted FFT coefficients into the original spatial–spectral feature domain through the inverse FFT (IFFT). Finally, we use a 1×1 convolutional layer to obtain the final result of SMA module. The complete fusion process is shown as follows:

$$\begin{aligned} A_o &= Fu\left(A_s, A_c\right) \\ &= Conv_{1\times 1}\left(\overline{IFFT}\left(\omega_1 FFT\left(A_s\right) + \omega_2 FFT\left(A_c\right)\right)\right) \end{aligned} \quad (9)$$

where $FFT(*)$ and $\overline{IFFT}(*)$ represent fast Fourier transform operation and inverse fast Fourier transform operation, respectively. ω_1 and ω_2 are learnable variables used to adaptively adjust the weights of A_c and A_s with the constraints of sum as one, so as to achieve the adaptive fusion of global and local features.

IV. Experimental Results and Analysis

A. Datasets and Metrics

To verify the performance of the proposed GAPT-Net in both quantitative and qualitative evaluations, we utilize the CAVE [40], NTIRE 2022 [41], and Harvard [42] datasets as training and evaluation benchmarks.

CAVE is a long-established and still commonly used hyperspectral dataset comprising various indoor objects with controlled lighting. The dataset was captured using tunable filters in 32 different scenes and contains 32 images with a spatial resolution of 512×512, with 31 consecutive spectral bands ranging from 400 to 700 nm in 10-nm increments.

The NTIRE 2022 dataset is the benchmark dataset for the NTIRE 2022 Spectral Reconstruction Challenge, a widely used benchmark for reconstructing HSIs from RGB images. The 2022 edition includes 950 RGB images and their corresponding HSIs. This open competition is divided into Clean Track and RealWorld Track. The Clean Track requires participants to reconstruct HSIs from noise-free RGB images and provides the spectral sensitivity function of the camera used to capture these RGB images. On the other hand, the RealWorld Track challenges participants to reconstruct HSIs from RGB images compressed as JPEG without providing the camera's spectral sensitivity function. Each image in this dataset has a spatial resolution of 482×512, with 31 spectral response values per pixel and wavelengths ranging from 400 to 700 nm. Without the camera's spectral response function, the reconstruction task is more complex, so we choose the RealWorld Track as the dataset for our experiments.

The Harvard dataset is widely recognized for its high spatial resolution and diverse real-world coverage, making it highly suitable for evaluating the robustness of reconstruction models under complex imaging conditions. This dataset consists of 50 HSIs, each containing 31 bands. The spectral range is 420–720 nm, with a spatial resolution of 1392×1040.

To comprehensively evaluate the performance of the proposed reconstruction model, we use the mean relative absolute error (MRAE), root-mean-square error (RMSE), peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and spectral angle mapper (SAM) as the evaluation metrics [43]. They are used to evaluate the accuracy of reconstructed results, where lower MRAE values, lower RMSE values, higher PSNR values, higher SSIM values, and lower SAM values indicate better reconstruction.

B. Experiment Setting

We use the PyTorch framework to train and test the reconstruction network on the CAVE, NTIRE 2022, and Harvard datasets. The NTIRE 2022 dataset uses 940 randomly selected images from the Real-world dataset as the training set and the remaining 10 images as the test set. Randomly selected 28 images from the CAVE dataset were used for training and the remaining 4 images were used for testing. Harvard

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TABLE I
COMPUTATIONAL EFFICIENCY OF MULTIPLE METHODS AND THEIR EVALUATION METRICS ON CAVE DATASET, NTIRE 2022 DATASET, AND HARVARD DATASET

Method	HSCNN+ [18]	AWAN [20]	HR-Net [44]	MST [29]	MS^{3}T [30]	MST++ [29]	SPECAT [31]	SSTHyper [35]	GAPT-Net
Params (M)	4.65	4.04	31.70	2.45	2.67	1.62	**0.32**	1.39	0.83
FLOPS (G)	304.45	270.61	163.81	32.07	25.40	23.05	14.08	**4.55**	16.76
CAVE Dataset									
MRAE	0.6706	0.4240	0.5226	0.4057	0.3777	0.3334	0.2895	0.2509	**0.2354**
RMSE	0.0592	0.0575	0.0564	0.0392	0.0311	0.0352	0.0270	0.0293	**0.0247**
PSNR	24.55	24.81	24.58	28.31	29.62	29.11	31.32	31.68	**32.15**
SSIM	0.9483	0.9528	0.9435	0.9677	0.9684	0.9717	0.9825	0.9823	**0.9839**
SAM	11.2783	9.5174	10.4984	8.0229	7.3645	7.4906	6.8703	7.0296	**6.4419**
NTIRE 2022 Dataset									
MRAE	0.3395	0.3610	0.3016	0.2831	0.2553	0.2358	0.2429	0.2394	**0.2265**
RMSE	0.0768	0.0945	0.0728	0.0551	0.0477	0.0461	0.0432	0.0461	**0.0431**
PSNR	24.69	23.49	25.06	27.04	28.46	28.74	29.08	29.28	**29.44**
SSIM	0.9348	0.9294	0.9417	0.9536	0.9529	0.9573	0.9619	0.9593	**0.9637**
SAM	12.7449	11.8546	9.4983	8.3898	8.7732	7.4362	7.0197	6.9804	**6.8359**
Harvard Dataset									
MRAE	0.2251	0.2218	0.2148	0.2032	0.2094	0.1949	0.1957	0.1959	**0.1926**
RMSE	0.0027	0.0028	0.0025	0.0021	0.0019	0.0017	0.0017	0.0018	**0.0016**
PSNR	27.03	27.46	28.05	30.18	29.85	30.58	30.63	30.14	**30.92**
SSIM	0.8982	0.9011	0.9142	0.9289	0.9248	0.9338	0.9359	0.9283	**0.9412**
SAM	10.8990	9.7384	9.8302	8.9032	8.7925	8.0583	7.9275	8.2106	**7.4853**

dataset uses 30 randomly selected images as the training set and the remaining 20 images as the test set. During the training process, each original image in the training dataset was sequentially cropped to an image of size 128 × 128 in terms of spatial dimensions and normalized to a range of 0–1. Therefore, for the 940 images in the NTIRE2022 training set, we obtained 11 280 128 × 128 image patches; for the 28 images in the CAVE training set, we obtained 448 128 × 128 image patches; and for the 30 images in the Harvard training set, we obtained 360 128 × 128 image patches. Meanwhile, simple data augmentation techniques such as random rotation and flipping were applied to the training set. For the test set, only cropping and data normalization were performed without any additional data augmentation techniques. Consequently, for the ten images in the NTIRE 2022 test set, we obtained 120 128 × 128 test images. For the four images in the CAVE test set, we obtained 64 128 × 128 test images. For the 20 images in the Harvard test set, we obtained 240 128 × 128 test images. The training process utilized the Adam optimizer to update model parameters, with detailed parameter settings, including $\beta_1 = 0.9$ and $\beta_2 = 0.999$. A total of 300 epochs were conducted during training, with a learning rate set to 0.0001 and a batch size of 20. Model training was conducted on 4 NVIDIA RTX 2080Ti (12 GB) GPUs, with hardware specifications including an AMD Ryzen 5 2600 @ 3.40 GHz processor and an Ubuntu 16.04 development environment.

C. Comparison With Other Methods

In this section, to verify the effectiveness of the proposed reconstruction network GAPT-Net, we compare it with multiple state-of-the-art methods, including three CNN-based methods (HSCNN++, AWAN, and HR-Net) and three Transformer-based methods (MST, MST++, MS^{3}T, SPECAT, and SSTHyper [35]). The comparison experiments are conducted on the CAVE, NTIRE 2022, and Harvard datasets.

1) Quantitative Results: MRAE, RMSE, PSNR, SSIM, and SAM are selected as the quantitative metrics to compare the reconstructed HSIs generated by different reconstruction networks with the ground-truth HSIs. Combining these metrics can effectively capture spectral and spatial fidelity. SAM emphasizes spectral consistency and SSIM focuses on structural similarity. Table I shows the metric values of reconstruction results of different methods on three datasets. The best results are highlighted in bold. From these results, we observe that compared to methods based on CNNs, Transformer-based methods have improved reconstruction quality, indicating the advantage of Transformers in representing spatial–spectral correlations. In addition, compared with other Transformer-based methods, our proposed GAPT-Net integrates global contextual information and local spatial–spectral details through SMA in GAP-Trans, effectively improving reconstruction quality with the lowest MRAE, SAM, and RMSE, and highest PSNR and SSIM. The effectiveness of the specific design in GAP-Trans will be further verified through ablation experiments in Section V.

Table I also lists the number of parameters and FLOPs of different network models. As our GAP-Trans calculates correlations along the spectral dimension of local patches, it effectively reduces computational complexity. Although the

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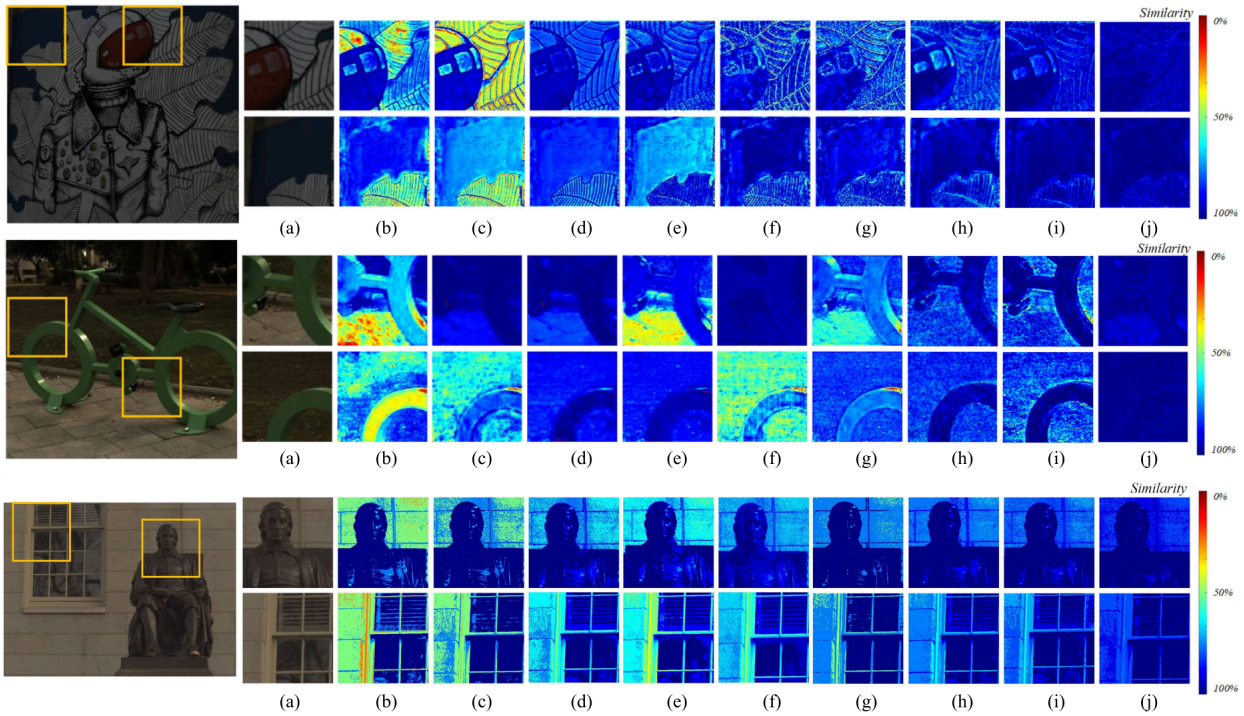


Fig. 5. Pseudo-color error map between HSIs reconstructed by different methods and the ground truth. The color bar indicates the degree of difference.(a) RGB. (b) HSCNN++. (c) AWAN. (d) HRNet. (e) MST. (f) MS^{3}T. (g) MST++. (h) SPECAT. (i) SSTHyper. (j) GAPT-Net.

FLOPs and parameters of our GAPT-Net are higher than those of SPECAT [31] and SSTHyper, our reconstruction performance is significantly better than theirs. Notably, SSTHyper emphasizes a lightweight design and adopts a cross-level fusion network to reduce computational cost. For the MST series, the entire spectral band is treated as tokens, and spectral-wise attention is computed globally. While this design enables the attention operation to be performed in a single pass over the whole image, its overall computational complexity is comparable to that of partitioning the input into multiple patches. However, in order to improve the reconstruction quality, MST cascades more Transformers at each level of the encoder and decoder than ours, thereby increasing their parameters and computational complexity.

2) Qualitative Results: In order to qualitatively assess the perceived quality of the reconstructed HSIs, Fig. 5 visualizes the HSIs reconstructed by different methods on selected bands, along with their corresponding error maps compared to the ground truth. From left to right, they correspond to HSCNN++, AWAN, HR-Net, MST, MS^{3}T, MST++, SPECAT, SSTHyper, and our GAPT-Net.

We observe that the error in the reconstructed HSI of GAPT-Net is significantly lower, indicating that it is closer to the true HSI. Additionally, thanks to the multiscale local patch reconstruction guided by global contextual information, the proposed network exhibits better detail recovery ability, maintains spatial smoothness in uniform regions, and avoids unpleasant anomalies and dust textures. Among the various CNN-based approaches, HR-Net performs relatively well. However, in some areas, there is a significant deviation between the reconstruction results of HR-Net and the ground truth. This deviation leads to its quantitative evaluation metrics failing to surpass those of Transformer-based methods. On the other hand, other CNN-based methods perform poorly, with larger outliers appearing in some regions.

TABLE II
COMPARISON OF EVALUATION METRICS OF FOUR ABLATION EXPERIMENTS AND GAPT-NET

Module	Abl. Meth.1	Abl. Meth. 2	Abl. Meth. 3	Abl. Meth. 4	GAPT-Net
(a) Traditional Spectral Trans	✓	✗	✗	✗	✗
(b) Local Spectral Trans	✗	✓	✓	✓	✓
(c) GTDG Augmentation	✗	✗	✓	✓	✓
(d) Spectral-wise Mixed Attention	✗	✗	✗	✓	✓
(e) Fast Fourier Fusion	✗	✗	✗	✗	✓
MRAE	0.3010	0.2415	0.2397	0.2370	**0.2265**

Despite some slight fluctuations in the reconstruction results, the overall quality of the Transformer-based method is still good, with no outliers present in the reconstruction process. GAPT-Net, on the other hand, benefits from the local reconstruction process guided by global contextual information and can leverage the advantages of Transformers in spectral reconstruction for multiscale patch images. As a result, it delivers relatively stable and high-quality reconstruction results.

We randomly select three HSIs of different scenes and plot the spectral signatures reconstructed by different methods along with the real spectral signatures at the marked spatial coordinates. Figs. 6– 8 illustrate the reconstructed

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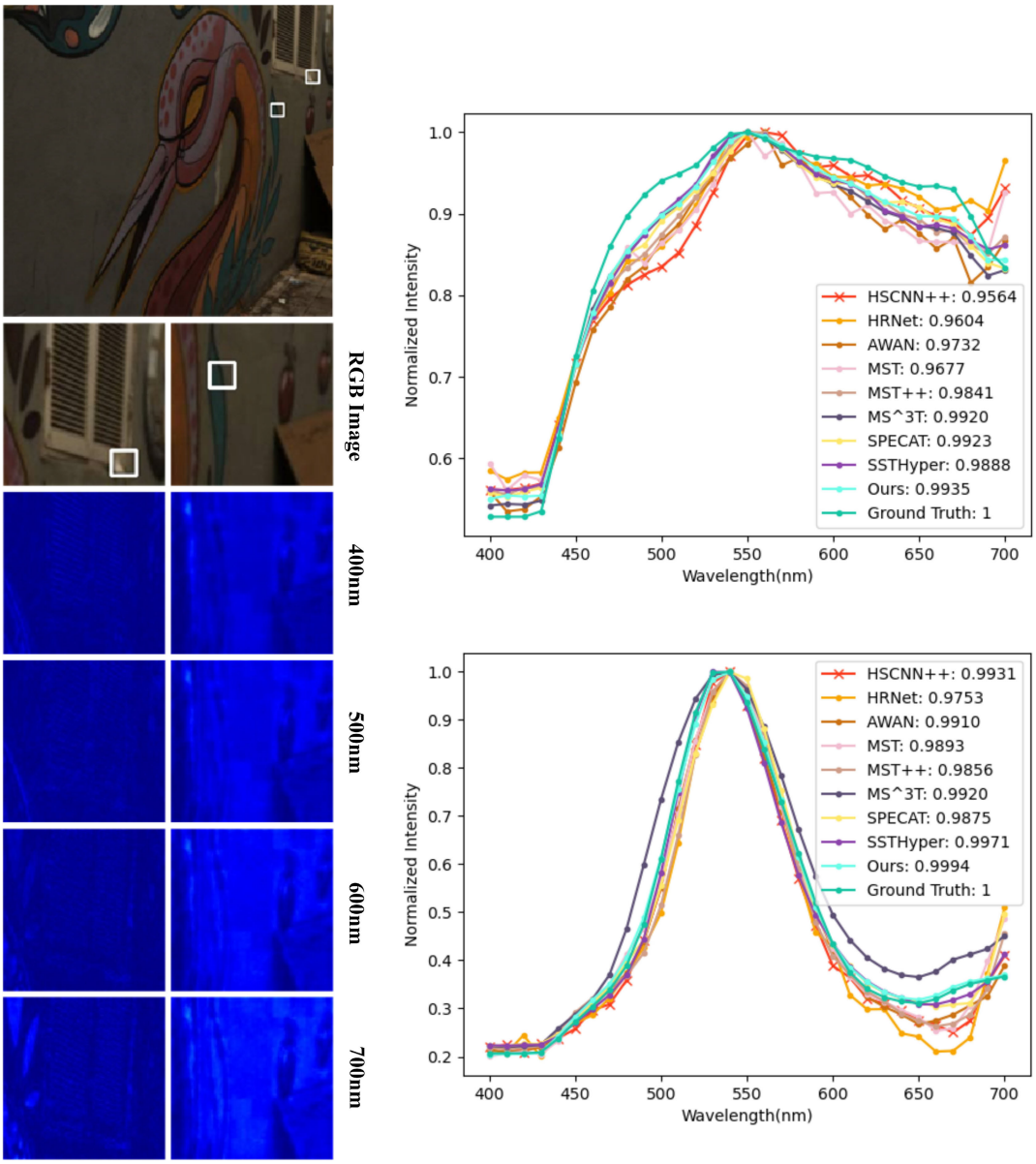


Fig. 6. Spectral signatures of the hyperspectral image (ARAD_1K_0913) from the NTIRE 2022 dataset reconstructed by different methods.

spectral signatures of different networks at the selected pixel points (white boxes) and the true spectral signature. Additionally, pseudo-color error maps between some reconstructed spectral bands by our GAPT-Net method and the ground-truth spectral band images are also provided, corresponding to the spectral bands including the 1st, 11th, 21st, and 31st bands. Based on the results presented in the line chart, it can be observed that in terms of the trend of spectral intensity values changing with the band, as well as the error between the reconstructed spectral intensity values and the real spectral intensity values of each band, GAPT-Net's reconstructions are more accurate compared to other methods. We also calculate the Pearson correlation coefficient to evaluate the accuracy of reconstructed spectral response signatures. The spectral signatures reconstructed by GAPT-Net have the highest accuracy. Additionally, from the pseudo-color error maps at the given bands, it can be observed that GAPT-Net can capture characteristic information of the spectral intensity values across the entire band and accurately fit the spectral intensity values on each band.

V. Ablation Studies

The proposed reconstruction network model mainly poses two novel designs: the global token dictionary generation module and the GAP-Trans. The core processing of GAP-Trans mainly consists of three components: a spectral-wise self-attention calculation mechanism based on local patches, a dual-branch mixed-attention processing, and a fast Fourier Fusion algorithm. By using a cascaded encoding–decoding pattern, the model generates high-quality HSIs. It effectively characterizes the correlation between global and local information, as well as the data coupling relationship between

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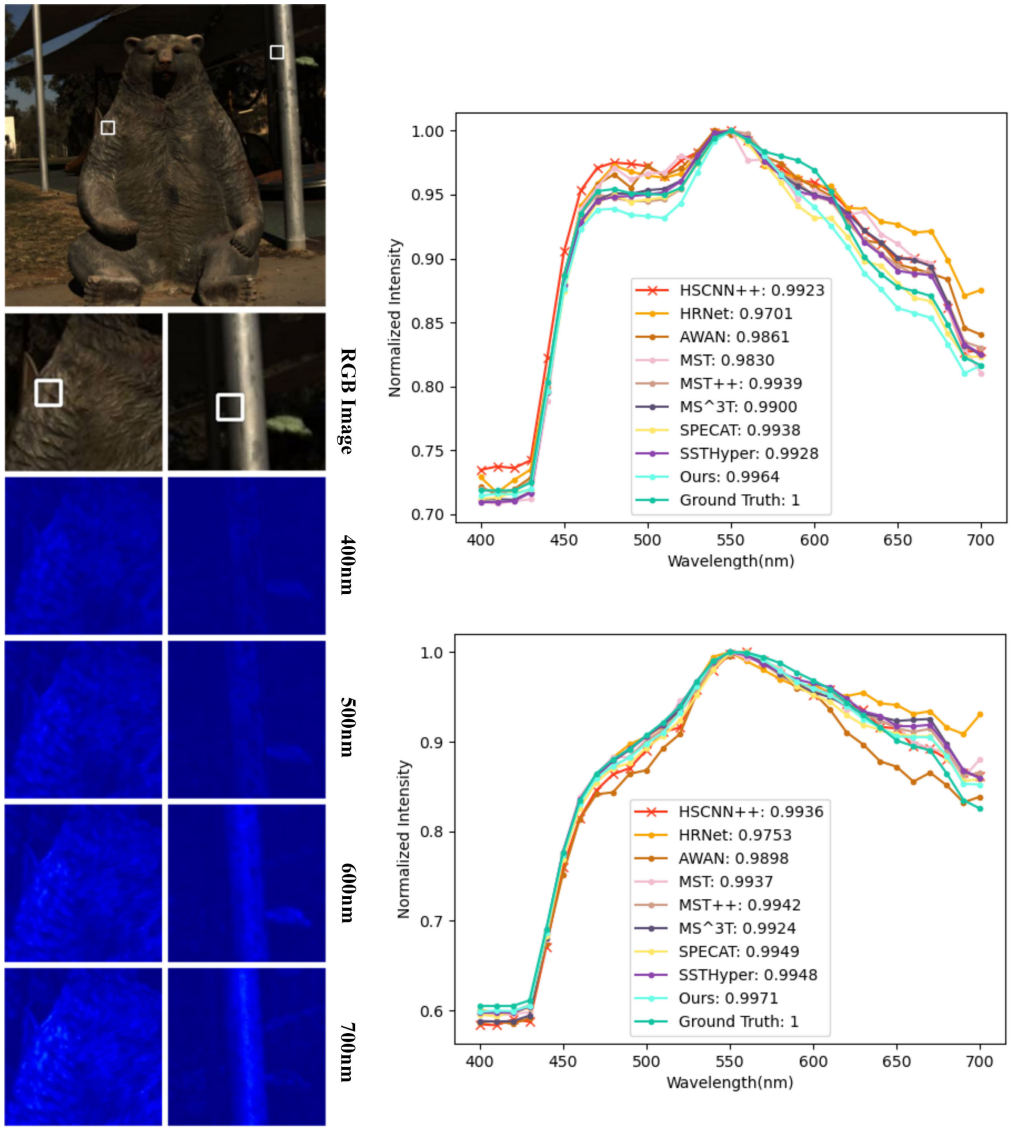


Fig. 7. Spectral signatures of the hyperspectral image(ARAD_1K_0915) from the NTIRE 2022 dataset reconstructed by different methods.

spatial and spectral dimensions, thus further improving the reconstruction quality.

In order to verify the validity of each part of the proposed reconstruction network GAPT-Net, we perform the ablation experiment by comparing GAPT-Net with four ablation methods. These four ablation methods are constructed by appropriately configuring five options: 1) traditional spectral dimension attention mechanism; 2) spectral dimension self-attention calculation mechanism based on local windows; 3) global token dictionary generation module; 4) dual-branch mixed-attention processing; and 5) fast Fourier fusion algorithm. The comparison experiments are conducted on the NTIRE 2022 test dataset.

The specific configuration of four ablation methods and their reconstruction results are shown in Table II, where the best results are highlighted in bold. Ablation method 1 based on the spectral dimension attention computation mechanism has the lowest reconstruction accuracy. Ablation method 2 shows that the spectral dimension self-attention calculation mechanism based on local windows can effectively guide the local self-attention calculation by introducing global contextual information and significantly improve the reconstruction quality. The comparison between ablation method 3 and ablation method 2 verifies the effectiveness of the global token dictionary generation module, which can provide more valuable global information. Its inclusion leads to improvements in reconstruction quality. The comparison of ablation method 4 with ablation methods 2 and 3 shows that the mode of mixed-attention computation can further improve the reconstruction quality of the model. This improvement is due to the effective fusion of global and local information for reconstruction. Finally, GAPT-Net uses the fast Fourier fusion algorithm to replace the simple addition operation for the two results obtained from the mixed-attention mode, which further improves the reconstruction quality of the model and achieves the lowest MRAE metric.

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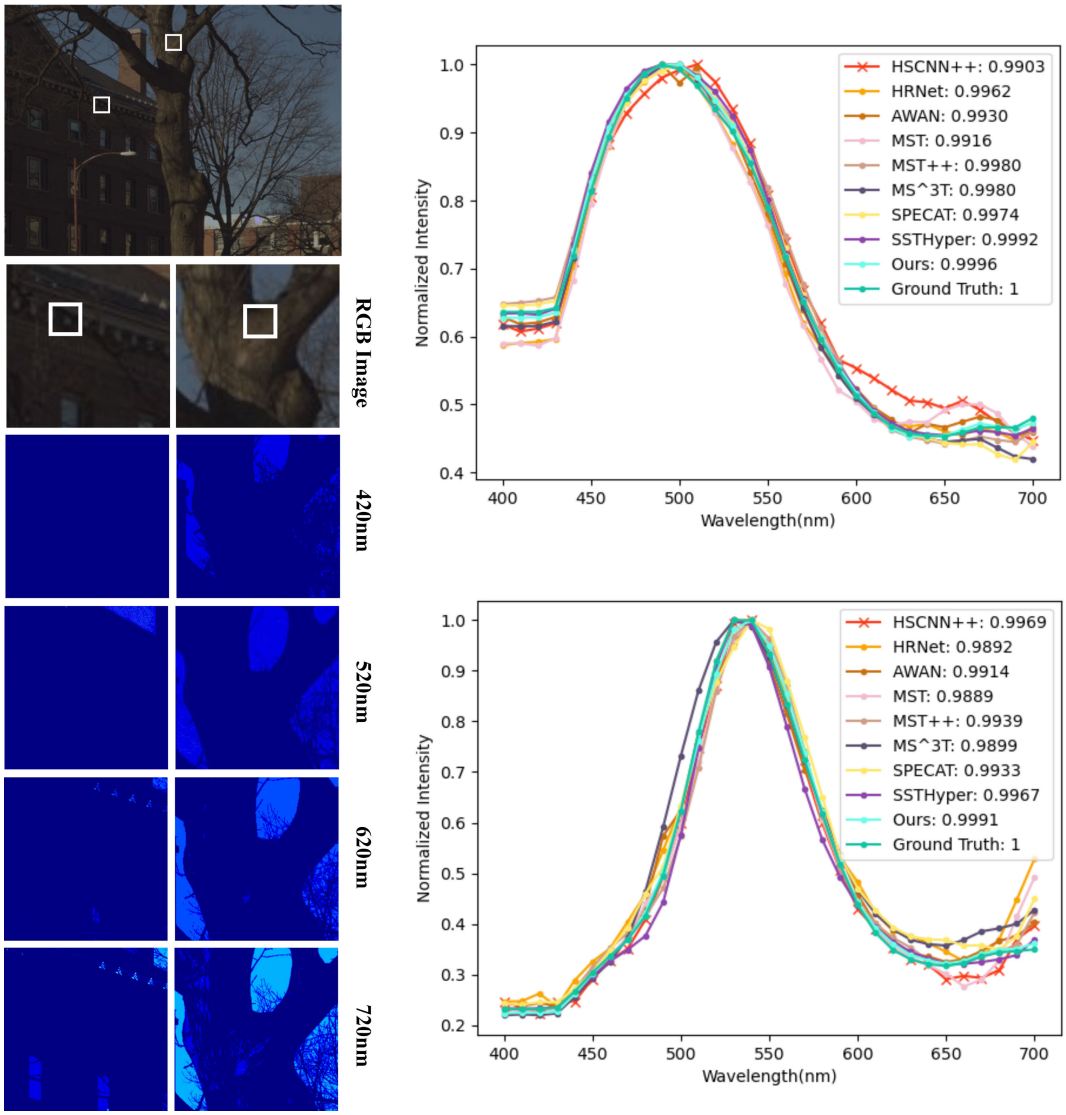


Fig. 8. Spectral signatures of the hyperspectral image from the Harvard dataset reconstructed by different methods.

VI. Conclusion

In this article, we propose a novel GAP-Trans for HSI reconstruction. The proposed network utilizes GAP-Trans as a basic component to extract features in both spatial and spectral dimensions and captures the coupled information between them. The mixed attention of GAP-Trans calculates the self-attention and cross-attention between the global token dictionary and the local patches, enabling the injection of global contextual information into each local window. This process boosts the fusion of global long-range context and local short-range fine details. Meanwhile, a Fast Fourier Fusion algorithm performs weighted fusion in the frequency domain on the results obtained from SMA, striking a balance between global and local features to further extract spatial and spectral features. Both quantitative and qualitative experiments have demonstrated that the reconstruction model proposed in this article can better restore HSIs with good spatial structure and accurate spectral features.

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